Higher National Diploma in Information Technology

Second Year, Second Semester Examination – 2023

HNDIT4212 : Machine Learning

Instructions for candidates:	No. of questions	: 06
Answer **five** questions **only**.	No. of pages	: 04
All questions carry **equal** marks.	Time	: Three (3) Hours

Question 01

1. Define the term Machine Learning?

(04 Marks)

2. Briefly explain followings:
   a. Supervised Learning
   b. Unsupervised Learning

(06 Marks)

3. What is **Discrete** random variable.

(04 Marks)

4. Consider the following terms used in Machine Learning. Write Most suitable words to given description. (Training, ML Algorithm, Dataset, Training Data, Testing Data, Accuracy)

   a) Data used to evaluate the performance of a trained model

   b) A method used to learn patterns from data

   c) The process of feeding data to the model to learn patterns

   d) The percentage of correct predictions made by the model

   e) A collection of structured data used in ML tasks

   f) Subset of data used to teach the model

(06 Marks)

[20 Marks]

Question 02

Consider the following training dataset that describes the weather conditions for the activity of **"Going for a Walk"**. The conditions include: **Sky** (Sunny, Cloudy, Rainy), **Temperature** (Hot, Mild, Cool), **Humidity** (High, Normal), **Windy** (TRUE, FALSE). The dataset has two classes: **C1: Going_for_a_Walk = Yes, C2: Going_for_a_Walk = No.**

Index	Sky	Temperature	Humidity	Windy	Go for Walk
0	Rainy	Hot	High	FALSE	No
1	Rainy	Hot	High	TRUE	No
2	Cloudy	Hot	High	FALSE	Yes
3	Sunny	Mild	High	FALSE	Yes
4	Sunny	Cool	Normal	FALSE	Yes
5	Sunny	Cool	Normal	TRUE	No
6	Cloudy	Cool	Normal	TRUE	Yes
7	Rainy	Mild	High	FALSE	No
8	Rainy	Cool	Normal	FALSE	Yes
9	Sunny	Mild	Normal	FALSE	Yes
10	Rainy	Mild	Normal	TRUE	Yes
11	Cloudy	Mild	High	TRUE	Yes
12	Cloudy	Hot	Normal	FALSE	Yes
13	Sunny	Mild	High	TRUE	No

You are given a new New Data: X =(**Sky = Rainy, Temperature = Hot, Humidity = High, Windy = FALSE).** You need to use the **Naïve Bayes classifier** to determine whether the person is likely to go for a walk. To find it calculate followings:

1) Compute P(C) The prior probability of each class can be computed based on the training tuples: P(Going_for_a_Walk =yes) and P(Going_for_a_Walk =no).

(04 Marks)

2) Compute P(X| Going_for_a_Walk =yes)

(06 Marks)

3) Compute P(X|Ci), P(X| Going_for_a_Walk =no)

(06 Marks)

4) Compute P(X|Ci)P(Ci) for each class and give the Naïve Bayesian Classifier prediction.

(04 Marks)

[20 Marks]

Question 03

Consider the following dataset used in a single-neuron network. Here, x1, x2, x3, x4 are normalized input attribute values, and w1, w2, w3, w4 are the corresponding weight values.

x1	x2	x3	x4
0.5	0.3	0.6	0.1

w1	w2	w3	w4
0.4	0.3	0.5	0.2

a) Compute the weighted sum for the given data

(04 Marks)

b) Use the following activation function to determine the output class:

$$f(x) = \begin{cases} 0 & \text{if } x < 0.5 \\ 1 & \text{if } x \geq 0.5 \end{cases}$$

Determine the **class label** for the given input.

(04 Marks)

2) You are required to normalize a dataset attribute B to the range [0,1]. The original minimum and maximum values of attribute B are 50 and 150 respectively. Use the min-max normalization formula:

$$v' = \frac{(v - \text{minB}) \times (\text{new_maxB} - \text{new_minB})}{\text{maxB} - \text{minB}} + \text{new_minB}$$

If the value of attribute B for a particular data point is v=90, calculate the normalized value v'.

(04 Marks)

3. Define the term Least Squares (Regression) Line.

(02 Marks)

4. Consider the data set related to Simple Linear Regression Line.

cov(X,Y)	-3421.34
S^2_X	18,930,340.00
$\bar{X}$	45,234.00
$\bar{Y}$	18,342.00

a) Calculate b_1

(03 Marks)

b) Calculate b_0

(03 Marks)

[20 Marks]

Question 04

1. Name two deep learning frameworks.

(04 Marks)

2. Explain difference between Machine Learning and Deep Learning.

(04 Marks)

3. Explain purpose of Optimizer in Deep Neural Network.

(06 Marks)

4. Briefly explain the ReLU Activation function used in deep learning.

(06 Marks)

[20 Marks]

Question 05

1. What is "Prediction Task" of data mining?

(04 Marks)

2. Classification is one of the common data mining tasks. Briefly explain it.

(04 Marks)

3. Compare Database Processing vs. Data Mining Processing

(06 Marks)

4. Principal component analysis is a variable reduction procedure. Explain it.

(06 Marks)

[20 Marks]

Question 06: Write short note on four topics.

1. Bayesian Networks
2. Components of a Monte Carlo simulation
3. Max Voting
4. Decision Nodes in Decision Tree
5. Gibbs Sampling

(05 Marks for each topic)

[20 Marks]